# Electronic Companion to “Robust Candidate MaxWeight Certificates for Heterogeneous Compute Scheduling: A Scheduleurm Case Study”

## EC. Conventional Proof Details

This appendix gives the conventional, paper-readable proof chain behind Theorem 1. It is intentionally redundant with the main text: the purpose is to make clear which part is mathematics, which part is a measured Scheduleurm certificate, and which part is checked by Lean. The Lean artifact is a formal check of the algebraic and recurrence implications; it is not a substitute for empirical certificates of service rows, lower confidence bounds, candidate admission, oracle gaps, or moment bounds.

### EC.1. Capacity Slack as a Support-Function Margin

Let $\mu(a) \in \mathbb{R}_+^{|\mathcal{I}|}$ denote the true mean service vector of full action $a \in \mathcal{A}^{\text{full}}$, and let $H_{\mathcal{A}^{\text{full}}}(q) = \max_{a\in\mathcal{A}^{\text{full}}} q^\top \mu(a)$ be the full-action support function for nonnegative backlog weights $q \in \mathbb{R}_+^{|\mathcal{I}|}$. A stationary randomized policy over full actions is a probability vector $x \in \mathrm{Mix}(\mathcal{A}^{\text{full}})$. If it has coordinate slack $\delta > 0$ against the arrival mean $\lambda$, then

$$\lambda_i + \delta \le \sum_{a\in\mathcal{A}^{\text{full}}} x_a \mu_i(a), \qquad i \in \mathcal{I}. \tag{1}$$

Multiplying each coordinate by $q_i \ge 0$ and summing gives

$$q^\top \lambda + \delta \,\|q\|_1 \le \sum_{a\in\mathcal{A}^{\text{full}}} x_a q^\top \mu(a). \tag{2}$$

Because $x$ is a convex combination, the right-hand side is at most the maximum full-action support:

$$\sum_{a\in\mathcal{A}^{\text{full}}} x_a q^\top \mu(a) \le H_{\mathcal{A}^{\text{full}}}(q). \tag{3}$$

Hence full-action capacity slack implies

$$q^\top \lambda + \delta \,\|q\|_1 \le H_{\mathcal{A}^{\text{full}}}(q). \tag{4}$$

This is the only place where the capacity-region hypothesis enters the drift proof. It converts an interior capacity statement into a MaxWeight support margin that can be compared to candidate actions. The paper uses the support-function statement as Assumption 1; the operational necessity direction is separate and requires a load-certified stochastic model and a conservation law.

## EC.2. Candidate Fabric Cover and Support Loss

The scheduler cannot enumerate every full configuration. It evaluates an admitted candidate family $\mathcal{A}_t^{\mathrm{cand}} \subseteq \mathcal{A}^{\mathrm{full}}$, possibly depending on the observed state $t$. The finite-candidate theorem needs a support loss certificate: for every $q \geq 0$,

$$H_{\mathcal{A}^{\mathrm{full}}}(q) \leq H_{\mathcal{A}_t^{\mathrm{cand}}}(q) + \epsilon_{\mathrm{cand}} \|q\|_1 . \tag{5}$$

When the loss is obtained from a calibrated fabric metric, let $\Phi(a)$ be the finite feature representation of action $a$, let $d_\Phi$ be the feature metric, and suppose each full action $a$ has a projected candidate $\pi_t(a) \in \mathcal{A}_t^{\mathrm{cand}}$ satisfying

$$d_\Phi(a, \pi_t(a)) \leq \rho. \tag{6}$$

If service is Lipschitz in this representation,

$$\|\mu(a) - \mu(\pi_t(a))\|_\infty \leq L\, d_\Phi(a, \pi_t(a)), \tag{7}$$

then for any maximizer $a^\star \in \arg\max_{a \in \mathcal{A}^{\mathrm{full}}} q^\top \mu(a)$,

$$\begin{aligned} H_{\mathcal{A}^{\mathrm{full}}}(q) &= q^\top \mu(a^\star) \\ &\leq q^\top \mu(\pi_t(a^\star)) + \|q\|_1 L\rho \\ &\leq H_{\mathcal{A}_t^{\mathrm{cand}}}(q) + L\rho \|q\|_1 . \end{aligned} \tag{8}$$

Thus $\epsilon_{\mathrm{cand}} = L\rho$ in the fabric-cover case. In measured finite slices where the candidate support is certified directly, the artifact uses the observed support gap itself as $\epsilon_{\mathrm{cand}}$; when the candidate set is exact on the measured slice, this term is zero. This is why the proof does not shrink the action space informally: every restriction must be paid for by the explicit support-function loss in (5).

Proposition 1 supplies a constructive version of this step. It forms candidates as representatives of active feature cells, bounds the candidate cardinality by the number of cells, and derives the same $L\rho$ support loss. Thus the theorem can consume either a directly audited support gap or a finite-cell construction certificate; only the latter supports a claim of nontrivial action-space compression.

## EC.3. Lower-Service Rows, Penalties, and Approximate Oracles

The robust scheduler optimizes conservative service, not the unknown true mean service. For each admitted candidate $a \in \mathcal{A}_t^{\mathrm{cand}}$, the artifact provides a lower-service vector $\mu_t^{\mathrm{L}}(a)$. The lower-service support loss is

$$H_{\mathcal{A}_t^{\mathrm{cand}}}(q) \leq \max_{a \in \mathcal{A}_t^{\mathrm{cand}}} q^\top \mu_t^{\mathrm{L}}(a) + \epsilon_{\mathrm{est}} \|q\|_1 . \tag{9}$$

The scheduling score is

$$\Psi_t(a) = Q^\top \mu_t^{\mathrm{L}}(a) - K_t(a) - G_t(a), \tag{10}$$

where $K_t(a)$ and $G_t(a)$ are bounded switching, risk, or guard penalties. Equation (9) is the lower-confidence-bound bridge from measured service rows to the drift proof. The coordinate-scaled sufficient condition for this bridge is given in Proposition 2: on the simultaneous confidence event it charges $2\epsilon_{\mathrm{LCB}} \sum_i s_i q_i$, or $2\epsilon_{\mathrm{LCB}} C_s \|q\|_1$ under a uniform scale bound. This prevents a high-throughput workload from defining the absolute error unit for all other classes. The selected action $a_t$ need not be the exact maximizer. It satisfies

$$\max_{a \in \mathcal{A}_t^{\mathrm{cand}}} \Psi_t(a) - \Psi_t(a_t) \le \alpha_0 + \alpha_1 \|Q\|_1\,, \tag{11}$$

and all admitted candidate penalties obey

$$0 \le K_t(a) + G_t(a) \le P_0 + \beta \|Q\|_1\,, \qquad a \in \mathcal{A}_t^{\mathrm{cand}}. \tag{12}$$

Combining the capacity support bound (4), the two support-loss bounds (5) and (9), the oracle gap (11), and the penalty bound (12) yields the selected lower-service margin. First,

$$\max_{a \in \mathcal{A}_t^{\mathrm{cand}}} Q^\top \mu_t^{\mathrm{L}}(a) \ge Q^\top \lambda + (\delta - \epsilon_{\mathrm{cand}} - \epsilon_{\mathrm{est}}) \|Q\|_1\,. \tag{13}$$

Second, by the oracle gap and the all-candidate penalty bound,

$$\begin{aligned} Q^\top \mu_t^{\mathrm{L}}(a_t) \ge{}& \max_{a \in \mathcal{A}_t^{\mathrm{cand}}} Q^\top \mu_t^{\mathrm{L}}(a) \\ &- (P_0 + \alpha_0) - (\beta + \alpha_1) \|Q\|_1 \\ \ge{}& Q^\top \lambda + \eta \|Q\|_1 - (P_0 + \alpha_0), \end{aligned} \tag{14}$$

where

$$\eta = \delta - (\epsilon_{\mathrm{cand}} + \epsilon_{\mathrm{est}} + \beta + \alpha_1). \tag{15}$$

The theorem therefore has a simple accounting rule: candidate loss, lower-service estimation loss, queue-scaled penalty, and queue-scaled oracle loss all consume the same capacity slack. A stability certificate requires $\eta > 0$. Additive terms $P_0$ and $\alpha_0$ do not change the slope of the drift; they enlarge the finite set outside which the drift is negative.

## EC.4. Quadratic Drift and the Foster Certificate

The queue update is

$$Q_i(t+1) = \left[Q_i(t) - S_i(t)\right]^+ + A_i(t), \tag{16}$$

where $A_i(t)$ is arrival and $S_i(t)$ is potential service for class $i$. Completed departure is $D_i(t) = \min\{Q_i(t), S_i(t)\}$. The displayed update remains valid because unused potential service is removed by the positive part. For $V(Q) = \frac{1}{2}\sum_i Q_i^2$, the standard one-step inequality gives

$$\begin{aligned} \Delta V(t) \le \frac{1}{2}\sum_i \big(A_i(t)^2 + S_i(t)^2\big) \\ + Q(t)^\top A(t) - Q(t)^\top S(t). \end{aligned} \tag{17}$$

Assumption 4 supplies the conditional second-order bound

$$\mathbb{E}\left[\frac{1}{2}\sum_i \big(A_i(t)^2 + S_i(t)^2\big) \,\middle|\, \mathcal{F}_t\right] \le B. \tag{18}$$

It also ties the stochastic process to the theorem-facing service rows:

$$\mathbb{E}[A(t) \mid \mathcal{F}_t] \le \lambda, \tag{19}$$

$$\mathbb{E}[S(t) \mid \mathcal{F}_t] \ge \mu_t^{\mathrm{L}}(a_t) \tag{20}$$

coordinatewise. Taking conditional expectation in (17) and using (14),

$$\begin{aligned} \mathbb{E}[\Delta V(t) \mid \mathcal{F}_t] &\le B + Q(t)^\top \lambda - Q(t)^\top \mu_t^{\mathrm{L}}(a_t) \\ &\le B + P_0 + \alpha_0 - \eta \, \|Q(t)\|_1 . \end{aligned} \tag{21}$$

If $N$ and $\gamma > 0$ satisfy

$$B + P_0 + \alpha_0 + \gamma \le \eta N, \tag{22}$$

then whenever $\|Q(t)\|_1 > N$, (21) implies

$$\mathbb{E}[\Delta V(t) \mid \mathcal{F}_t] \le -\gamma. \tag{23}$$

This is the finite-set drift certificate used in Theorem 1. To translate the certificate into positive recurrence for a Markov chain, the paper also assumes a countable augmented state space, a finite nonempty backlog sublevel set, and a closed communicating class or equivalent local-return condition. Without those recurrence-side conditions the theorem should be read as a negative-drift certificate, not as an unconditional positive-recurrence statement for every possible implementation.

## EC.5. Statewise Feasible Families and Operational Boundaries

The statewise corollary is a conditioning argument rather than a new drift identity. Let $\chi(t)$ record scheduler-visible information such as alive nodes, pinned tasks, queued jobs, and measured admission status. If the full family, candidate family, true and lower-service maps, feature map, penalty bounds, oracle bounds, and moment bounds are indexed by $X(t) = (Q(t), \chi(t))$ and satisfy the same constants for every admitted state, then the proof in EC.1–EC.4 holds after conditioning on $\mathcal{F}_t$. Uniform constants preserve both the drift and finite-horizon backlog bounds. Positive recurrence additionally requires the augmented-state conditions in Assumption 5; it does not follow merely by averaging over $\chi(t)$.

The operational capacity statement is kept separate. Sufficiency follows from the positive-slack drift chain above once a stochastic model certifies the arrival mean, service accounting, lower-service domination, moment bounds, and local recurrence conditions. Necessity is not obtained by defining a model to "stabilize" a load; it requires the model to encode the load and obey a conservation law showing that long-run departures cannot exceed the capacity base. This separation is essential for the paper's claim boundary: Lean checks the implication from certified assumptions to drift and recurrence, while the experiments and admission checks certify when Scheduleurm rows satisfy those assumptions.

## EC.6. Variable-Duration Configuration Trajectories

The slotted theorem does not silently cover migration, machine reconfiguration, or port moves whose durations differ. We therefore state the separate frame result used by the trajectory interface. Let $n$ index decision frames, let $\mathcal{F}_n^{\mathrm{F}}$ be the information available at the start of frame $n$, and let the $\mathcal{F}_n^{\mathrm{F}}$-measurable duration satisfy

$$0 < \tau_{\min} \le \tau_n \le \tau_{\max} < \infty. \tag{24}$$

The frame-boundary queue obeys

$$Q^{\mathrm{F}}(n+1) = \left[Q^{\mathrm{F}}(n) - S^{\mathrm{F}}(n)\right]^+ + A^{\mathrm{F}}(n), \tag{25}$$

where $A^{\mathrm{F}}(n)$ and $S^{\mathrm{F}}(n)$ are cumulative arrivals and potential service during the frame. Define

$$\overline{S}^{\mathrm{F}}(n) = \mathbb{E}\left[S^{\mathrm{F}}(n) \mid \mathcal{F}_n^{\mathrm{F}}\right]. \tag{26}$$

Suppose the conditional arrival and second-order bounds are

$$\mathbb{E}\left[A_i^{\mathrm{F}}(n) \mid \mathcal{F}_n^{\mathrm{F}}\right] \le \tau_n \lambda_i, \tag{27}$$

$$\mathbb{E}\left[\frac{1}{2}\sum_i \left\{(A_i^{\mathrm{F}}(n))^2 + (S_i^{\mathrm{F}}(n))^2\right\} \middle| \mathcal{F}_n^{\mathrm{F}}\right] \le B_{\mathrm{F}}. \tag{28}$$

Let $\epsilon_{\mathrm{F}}$ combine candidate-support and conservative-service losses in rate units. The cumulative approximate-oracle and penalty obligations are

$$\begin{aligned}\tau_n H_{\mathcal{A}^{\mathrm{full}}}\left(Q^{\mathrm{F}}(n)\right) \le \left(Q^{\mathrm{F}}(n)\right)^{\top} \overline{S}^{\mathrm{F}}(n) \\ + \tau_n(\epsilon_{\mathrm{F}} + \alpha_1)\left\|Q^{\mathrm{F}}(n)\right\|_1 \\ + \alpha_0 + P_n^{\mathrm{F}}.\end{aligned} \tag{29}$$

$$P_n^{\mathrm{F}} \le P_0 + \tau_n \beta \left\|Q^{\mathrm{F}}(n)\right\|_1. \tag{30}$$

All quantities in (29)–(30) are in cumulative work units over the same frame; in particular, migration lost work cannot be inserted as an unmatched wall-clock scalar. For the controlled migration rows, let $W_n^{\mathrm{rem}} > 0$ be one task's remaining work, let $\underline{\mu}_n^{\mathrm{old}}$ and $\underline{\mu}_n^{\mathrm{new}}$ be its per-task lower-service rates in the measured source and destination states, and let $T_n^{\mathrm{mig}}$ be the measured checkpoint, synchronization, staging, warmup, lost-work, and time-equivalent risk overhead. The migration trajectory is represented by the duration-normalized lower service

$$\underline{\mu}_n^{\mathrm{mig,eff}} = \frac{W_n^{\mathrm{rem}}}{T_n^{\mathrm{mig}} + W_n^{\mathrm{rem}}/\underline{\mu}_n^{\mathrm{new}}}. \tag{31}$$

It enters the candidate family only when

$$\frac{W_n^{\mathrm{rem}}}{\underline{\mu}_n^{\mathrm{old}}} - \frac{W_n^{\mathrm{rem}}}{\underline{\mu}_n^{\mathrm{new}}} > T_n^{\mathrm{mig}}, \tag{32}$$

equivalently $\underline{\mu}_n^{\mathrm{mig,eff}} > \underline{\mu}_n^{\mathrm{old}}$. Thus wall-clock overhead first changes the trajectory duration; it is not subtracted directly from a service-rate score.

For the detailed proof of Proposition 3, suppose (24)–(30) hold and the full family has support slack $\delta$. Define

$$\eta_{\mathrm{F}} = \delta - \epsilon_{\mathrm{F}} - \alpha_1 - \beta. \tag{33}$$

Then

$$
\begin{aligned}
&\mathbb{E}\Big[V\Big(Q^{\mathrm{F}}(n+1)\Big) - V\Big(Q^{\mathrm{F}}(n)\Big)\Big|\mathcal{F}_n^{\mathrm{F}}\Big] \\
&\quad \le B_{\mathrm{F}} + \alpha_0 + P_0 - \tau_n \eta_{\mathrm{F}} \left\|Q^{\mathrm{F}}(n)\right\|_1 \\
&\quad \le B_{\mathrm{F}} + \alpha_0 + P_0 - \tau_{\min} \eta_{\mathrm{F}} \left\|Q^{\mathrm{F}}(n)\right\|_1 ,
\end{aligned}
\tag{34}
$$

provided $\eta_{\mathrm{F}} > 0$. If the frame-boundary augmented state is countable, its backlog sublevel sets are finite and nonempty, and the usual local-return condition holds, then any $N \in \mathbb{N}$ and $\gamma > 0$ satisfying

$$
B_{\mathrm{F}} + \alpha_0 + P_0 + \gamma \le \tau_{\min} \eta_{\mathrm{F}} N \tag{35}
$$

give a finite-set Foster recurrence certificate for the embedded frame-boundary chain. Moreover, elapsed physical time through frame $n$ is at most $n\tau_{\max}$.

*Detailed proof of Proposition 3.* The quadratic queue inequality applied to (25), followed by conditional expectation, gives

$$
\begin{aligned}
\mathbb{E}[\Delta_{\mathrm{F}} V(n) \mid \mathcal{F}_n^{\mathrm{F}}] \le B_{\mathrm{F}} + \tau_n \Big(Q^{\mathrm{F}}(n)\Big)^{\top} \lambda \\
- \Big(Q^{\mathrm{F}}(n)\Big)^{\top} \overline{S}^{\mathrm{F}}(n),
\end{aligned}
\tag{36}
$$

where $\Delta_{\mathrm{F}} V(n)$ denotes the frame-to-frame Lyapunov increment. Full-action support slack implies

$$
\Big(Q^{\mathrm{F}}(n)\Big)^{\top} \lambda + \delta \left\|Q^{\mathrm{F}}(n)\right\|_1 \le H_{\mathcal{A}^{\mathrm{full}}}\Big(Q^{\mathrm{F}}(n)\Big). \tag{37}
$$

Multiplying (37) by $\tau_n$, substituting (29) and (30) into (36), and collecting coefficients yields the first inequality in (34). Positivity of $\eta_{\mathrm{F}}$ and $\tau_n \ge \tau_{\min}$ gives the second. Equation (35) makes the conditional drift at most $-\gamma$ outside the finite sublevel set, so the same local-return argument as in EC.4 applies to the embedded chain. Finally, $\sum_{k=0}^{n-1} \tau_k \le n\tau_{\max}$ proves the calendar-time bound.

If arrivals and service are rate-scaled throughout a frame and their unit-time second-order term is at most $B$, then the formal artifact also checks the constructive choice $B_{\mathrm{F}} \le \tau_{\max}^2 B$. Setting $\tau_n = \tau_{\min} = \tau_{\max} = 1$, $B_{\mathrm{F}} = B$, and $\epsilon_{\mathrm{F}} = \epsilon_{\mathrm{cand}} + \epsilon_{\mathrm{est}}$ recovers the slotted drift theorem. Thus the result strengthens rather than relabels the migration boundary: a measured migration row must still certify cumulative net service, duration, checkpoint/synchronization/resume loss, and risk before it can enter a stochastic frame model.

## EC.7. Lean Crosswalk and Artifact Assumptions

Tables 1 and 2 separate the formal proof contract from the empirical Scheduleurm certificates. The manuscript uses paper-facing statement names; exact Lean identifiers are kept in the supplementary theorem crosswalk so that the article does not become a code index.

**Table 1** Paper statements and Lean proof groups.

| Paper statement | Lean proof group | What the formal result checks |
|---|---|---|
| Proposition 1 | Finite feature-cell candidate construction theorem | Candidate containment, cardinality bounded by active feature cells, and nonzero $L\rho$ support loss. |
| Theorem 1, exact candidate version | Fixed-family exact-oracle robust candidate MaxWeight theorem | Support-function candidate loss, robust lower-service drift, and finite-set recurrence certificate. |
| Theorem 1, approximate-oracle version | Fixed-family approximate-oracle robust candidate MaxWeight theorem | Calibrated fabric cover, bounded second moment, additive and queue-scaled oracle loss, and slack accounting. |
| Corollary 2 | State-indexed candidate-family theorem | State-dependent feasible families under the same uniform constants used in the fixed-family theorem. |
| Proposition 2 | Diagonal-scaled lower confidence bound theorem | Heterogeneous service-unit scaling and lower-service support loss in normalized coordinates. |
| Corollary 1 | Foster telescoping backlog theorem | Finite-horizon cumulative-backlog and asymptotic expected-backlog bounds from the residual drift margin. |
| Proposition 3 | Variable-duration frame drift and embedded recurrence theorems | Cumulative-work oracle, bounded frame penalty, uniform frame drift, duration-one reduction, and elapsed-time bound. |
| Operational capacity boundary | Load-certified capacity sandwich theorem | Positive-slack sufficiency and zero-slack conservation-law necessity once the stochastic model encodes arrivals and service accounting. |

The exact Lean theorem identifiers, file hashes, and build log are supplied in the supplementary artifact crosswalk. This table records the mathematical correspondence used by the paper.

**Table 2** Formal hypotheses and Scheduleurm certificates.

| Object | Formal role | Scheduleurm certificate or boundary |
|---|---|---|
| $\lambda$, $\delta$ | Arrival mean and full-action capacity slack. | Online arrival replay, load sweep, and finite lower-service capacity checks. |
| $\epsilon_{\text{cand}}$ or $L\rho$ | Candidate-set or fabric-cover support loss. | Measured finite-slice support gaps, declared bucket admission, and fabric-cover calibration rows when available. |
| $\mu_t^{\text{L}}$, $\epsilon_{\text{est}}$ | Conservative lower-service rows and estimation loss. | Holdout lower confidence bound checks, measured service cache, and admission/probe status. |
| $P_0, \beta$ | Bounded and queue-scaled penalty consumption. | Penalty-unit records, guard/tie-break checks, and bounded switching configuration. |
| $\alpha_0$, $\alpha_1$ | Additive and queue-scaled approximate oracle loss. | Oracle-gap check comparing selected action against theorem-facing candidate families. |
| $B, N, \gamma$ | Second-order moment bound and finite-set drift threshold. | Workload bounds, controlled trace statistics, and recurrence-side finite-state admission conditions. |
| $\tau_{\min}, \tau_{\max}, B_{\text{F}}$ | Frame-duration bounds and cumulative second-order term. | A trajectory row must use natural elapsed duration and cumulative arrivals/service; unit-rate scaling yields the audited bound $B_{\text{F}} \leq \tau_{\max}^2 B$ when applicable. |
| $\overline{S}^{\text{F}}(n), P_n^{\text{F}}$ | Conditional cumulative service and frame-level migration/reconfiguration loss. | Checkpoint flush, synchronization, staging, resume warmup, lost work, and failure risk must be converted to the same cumulative service/penalty units; deterministic domain holdouts alone do not certify this stochastic bridge. |
| Statewise $\chi(t)$ | Scheduler-visible feasible-family index. | Alive node, pinned-job, queued-job, service-cache, and admission-state trace fields. |
| Operational model certificate | Links stochastic arrivals and services to the queueing theorem. | Load-certified model contract plus conservation-law boundary; not inferred from Lean alone. |

The table is deliberately conservative: Lean verifies the implications once these hypotheses hold, while the Scheduleurm artifact determines which measured rows satisfy them.

## Appendix: Supplementary Evidence Index

The full machine-readable status dashboard is provided in the supplementary artifact package. Tables 3 and 4 give a compact index of what is included and what each evidence family is allowed to support. They summarize the detailed supplementary status dashboard in manuscript-facing evidence categories; the row-level dashboard remains in the supplement, so the compression changes presentation rather than the underlying evidence.

**Table 3** Supplementary evidence index: theorem and Scheduleurm certificates.

| Gate family | Retained dashboard evidence | Boundary kept in manuscript |
|---|---|---|
| Formal proof package | Consolidated Lean upload, toolchain, build log, theorem crosswalk, no-`sorry`/`admit`/`axiom` proof search, and checksums. | Formalizes theorem implications only after the stated service, moment, slack, and recurrence assumptions are certified. |
| Declared finite-domain cover | q00/q01/q10/q11 service-cache summaries, admission boundaries, measured support gaps, all-state safety cover, and probe/defer status for unknown states. | Supports declared finite measured domains; it does not assert positive service for arbitrary future states. |
| Lower-service and slack certificates | Holdout lower confidence bound (LCB) rows, selected-profile stochastic LCB certificate, row-level lower-service slack, $L\rho$ accounting, and finite second-moment thresholds. | Certifies lower-service capacity for admitted rows; mean-service and broad fabric-cover claims require additional calibration. |
| Replay and online stress | Explicit task-list replay, Poisson and bursty load sweeps, ablations, guarded-policy comparisons, and slack accounting tables. | Compares policies on the same measured service cache; it is not a proof that a static replay trace is a stationary arrival process. |
| Controlled launch/completion | Six-task and 32-task theorem-admitted launches, completion records, candidate rows, and $\alpha_0 = \alpha_1 = 0$ oracle-gap checks. | Supports controlled hook execution; it is separate from production-wide organic dispatch claims. |
| Production-history bridge | Strict completed-active production classification, strict launched/completed counts, read-only shadow traces, service-map oracle bridge, and exclusion of attempted-only or external rows. | Supports admitted history evidence; raw telemetry and future unknown jobs are outside the theorem population. |
| Live-oracle and global-dispatch readiness | Optional global-batch soft-hint hook, live-oracle trace fields, active-production shadow rows, and production readiness bridge. | Shows hook readiness and read-only conversion to theorem rows; it does not make the global batch policy the default production dispatcher. |

This table indexes evidence used for the main theorem, measured finite-slice claims, and Scheduleurm production-history bridge.

**Table 4** Supplementary evidence index: external-policy and extension checks.

| Gate family | Retained dashboard evidence | Boundary kept in manuscript |
|---|---|---|
| Policy-semantics baseline universe | Gavel-, Pollux-, Sia-, IADeep-, Salus-, fairness-, packing-, short-job, and delay-style policy rows evaluated on the same measured service cache. | Tests whether the candidate family omits useful finite scheduling semantics; it is not direct binary/full-stack superiority. |
| Gavel calibration and holdout rows | Adapter certificate, profile-aware calibration, resident-delay/JCT holdout, native same-workload simulator row, and same-host physical probe. | Keeps scalar service-unit equivalence separate from measured-cache policy comparison and scoped runtime evidence. |
| Named external runtime probes | Same-host runtime or adapter probes for Gavel, Pollux/AdaptDL, Sia, IADeep, and Salus, with paired workload and environment metadata when available. | Diagnostic implementation evidence only; original multi-node control-plane superiority is not claimed. |
| External action-union and algorithm upgrades | External policy-family actions, trajectory/action-semantics union rows, adaptive scalarized replay, state-dependent marginal lookup, global lookahead, and ETA reuse. | Shows the candidate family can absorb registered semantics; it is not production-default launch evidence. |
| Controlled migration-cost probes | Physical checkpoint, sync, and resume timing rows for controlled CNN, hybrid-RL, and CPU-surrogate payloads at 25%, 50%, and 75% progress. | Calibrates $K_{\mathrm{mig}}$ for benchmark migration actions only; ordinary running user jobs remain excluded. |
| Adjacent-domain scheduler bridge | Decima-style Spark-DAG simulator bridge, registered non-GPU scheduling diagnostics, and same-domain simulator notes. | Shows adjacent-domain coverage; it is not a GPU co-location full-stack comparison. |
| Learning and hidden-regime extensions | Active-bucket concentration, forced exploration, dwell/switching, detector-delay, optional live-hook records, and regime-detector fields. | Extension evidence; the current default live scheduler does not claim a complete online-learning theorem. |
| Non-future closure and future-state safety | Conservative all-state safety cover, probe/defer admission, zero theorem service for unknown states, and non-future closure rows excluding arbitrary future workloads. | Prevents overclaiming positive service or superiority for arbitrary future workloads, unregistered schedulers, or unmeasured hardware states. |

This table keeps external scheduler and extension evidence visible while preserving the paper's scoped mathematical claim.

## Code and Data Availability

The code, experiment artifacts, measured service-cache summaries, reviewer supplement manifest, and README files for this submission are provided in the supplementary Scheduleurm artifact package. The package contains the Scheduleurm source tree, theorem-facing replay modules, measured service-cache files, generated experiment summaries, and the Lean supplement headed by `ScheduleurmUpload.lean`. The reproduction script rebuilds the paper PDF, reruns the safe replay and certificate checks, executes the targeted tests, and refreshes the reproduction manifest. Live launches and external full-stack runtime probes are not executed by default; their recorded outputs are packaged with the corresponding environment notes. Raw scheduler history records that are outside the theorem-facing completed-active population are retained as operational telemetry and are not used as the claimed replication population. The Lean supplement package contains `ScheduleurmUpload.lean`, `lakefile.toml`, `lean-toolchain`, `build.log`, `theorem_crosswalk.md`, `manifest.json`, and checksums.